

UPPER BOUND OF MINIMAL LOG DISCREPANCIES

JIAO LIU

ABSTRACT. We show that the minimal log discrepancy of any variety of fixed dimension has an upper bound. Our result based on the recent result of OpenAI [OAI26, Proposition 2.1].

We work over the field of complex numbers $\mathbb{C}$. The goal of this paper is to prove the conjecture on the upper bound of mlds, provided that [OAI26, Proposition 2.1] is correct.

Theorem 0.1. *Let d be a positive integer. Then there exists a positive integer $N = N(d)$ depending only on d satisfying the following: for any d -dimension pair (X, B) and any $x \in X$, we have*

$$\text{mld}(X \ni x, B) \leq N(d).$$

Proof. Let $f : Y \rightarrow X$ be a $\mathbb{Q}$ -factorial dlt modification of (X, B) and let $B_Y := f_*^{-1}(B \wedge \text{Supp } B) + \text{Exc}(f)$. Possibly replacing (X, B) with (Y, B_Y) , we may assume that (X, B) is $\mathbb{Q}$ -factorial dlt. Possibly replacing B with 0, we may assume that (X, B) is $\mathbb{Q}$ -factorial klt. Possibly replacing (X, B) with a $\mathbb{Q}$ -factorial terminalization, we may assume that X is terminal. Possibly replacing B with 0, we may assume that $B = 0$ and X is $\mathbb{Q}$ -factorial terminal. Possibly cutting X by hyperplane sections and induction on dimension, we only need to show that for any closed point $x \in X$, $\text{mld}(X \ni x) \leq N(d)$ for some $N(d)$. In the following we fix such a closed point x .

Let $h : W \rightarrow X$ be a resolution of X such that

$$\mathfrak{m}_x \mathcal{O}_W = \mathcal{O}_W(-F), \quad F = \sum b_i F_i$$

where $F_i \geq 0$ and $\text{Supp } F = h^{-1}(x)$. Let

$$a_i := a(F_i, X) > 1$$

for each i and let $t := \min_i \frac{a_i - \frac{1}{2}}{b_i}$. Possibly switching indices, we may assume that $t = \frac{a_1 - \frac{1}{2}}{b_1}$. Let $E := F_1$, then we have $a_i - tb_i \geq \frac{1}{2}$ for any i and $a(E, X) = \frac{1}{2} + t \text{mult}_E \mathfrak{m}_x$. By Skoda's theorem, we have

$$t < \min_i \frac{a_i}{b_i} = \text{lct}(X, \mathfrak{m}_x) \leq d,$$

hence $0 < t < d$. Possibly shrinking X to an affine neighborhood of x and pick general elements $H_1, \dots, H_{d+1} \in |\mathfrak{m}_x|$, we let $\Delta := \frac{t}{d+1} \sum_{i=1}^{d+1} H_i$ and consider (X, Δ) . Then we have $h^* H_i = F + h_*^{-1} H_i$ for any i . Since H_i are general, $(W, \text{Exc}(h) + \sum_{i=1}^{d+1} h_*^{-1} H_i)$ is log smooth. For any h -exceptional prime divisor D , we have $a(D, X, \Delta) = a(D, X) > 1$ if $\text{center}_X D \neq x$ and

$$a(F_i, X, \Delta) = a(F_i, X) - t \text{mult}_{F_i} F = a_i - tb_i = \frac{1}{2} + (a_i - \frac{1}{2}) - tb_i \geq \frac{1}{2}.$$

Date: October 9, 2026.

2020 Mathematics Subject Classification. 14E30, 14B05.

Key words and phrases. Minimal log discrepancy. Upper bound.

for any i . In particular,

$$a(E, X, \Delta) = a_1 - \frac{a_1 - \frac{1}{2}b_1}{b_1}b_1 = \frac{1}{2}.$$

By [BCHM10, Corollary 1.4.3], there exists an extraction $g : Z \rightarrow X$ of E . Let $E_Z := \text{center}_Z E$. Since X is $\mathbb{Q}$ -factorial, Z is $\mathbb{Q}$ -factorial and $-E_Z$ is ample/ X . Thus K_Z is ample/ X . We have that (X, Δ) is $\frac{1}{d+1}$ -lc, so $(Z, \frac{1}{2}E + g_*^{-1}\Delta)$ is $\frac{1}{d+1}$ -lc, and so Z is $\frac{1}{d+1}$ -lc. By [OAI26, Proposition 2.1], there exists a real number $c = c(d, \frac{1}{d+1})$ depending only on d , such that

$$\text{lct}(Z, \mathfrak{m}_x \mathcal{O}_Z) \geq c(d, \frac{1}{d+1}).$$

This implies that $c(d, \frac{1}{d+1}) \text{mult}_E \mathfrak{m}_X \leq 1$, hence $\text{mult}_E \mathfrak{m}_x \leq \frac{1}{c(d, \frac{1}{d+1})}$. We have

$$\text{mld}(X) \leq a(E, X) = a(E, X, \Delta) + \text{mult}_E \Delta = \frac{1}{2} + \frac{t}{d+1} \cdot (d+1) \text{mult}_E \mathfrak{m}_x < \frac{1}{2} + \frac{d}{c(d, \frac{1}{d+1})}.$$

The theorem follows. $\square$

Acknowledgements. This work was supported by the National Key R&D Program of China (Grant No. 2024YFA1014400). The author thanks Caucher Birkar for useful discussions. No AI is necessary for the production of this paper.

REFERENCES

- [BCHM10] C. Birkar, P. Cascini, C. D. Hacon, and J. McKernan, *Existence of minimal models for varieties of log general type*, J. Amer. Math. Soc. **23** (2010), no. 2, 405–468.
- [OAI26] OpenAI, *Bounded klt complements for Fano contractions*, <https://github.com/openai/math/blob/main/preprints/Log-abundance-in-characteristic-zero-September-24-2026/paper.pdf>.

DEPARTMENT OF MATHEMATICS, PEKING UNIVERSITY, NO. 5 YIHEYUAN ROAD, HAIDIAN DISTRICT, BEIJING 100871, CHINA

BEIJING INTERNATIONAL CENTER FOR MATHEMATICAL RESEARCH, PEKING UNIVERSITY, NO. 5 YIHEYUAN ROAD, HAIDIAN DISTRICT, BEIJING 100871, CHINA

Email address: liujihao@math.pku.edu.cn